



SECURITY MONITORING



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LAB REPORT 1 R00221306

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Lab 1

First Attempt completed tasks 1-4

4 OF 5	35M 14s	220 OF 240 (91.67%)
OBJECTIVES COMPLETE	ELAPSED TIME	SCORE
WEDNESDAY, 18 SEPTEMBER 2024, WEDNESDAY, 18 SEPTEMBER 2024,		
11:17 AM	11:53 AM	
START TIME	END TIME	
OBJECTIVE	COMPLETED	SCORE
LAUNCH AND DETECT A TCP SYN FLOOD ATTACK	✓	55 OF 55 (100.00%)
MITIGATE A TCP SYN FLOOD ATTACK	✓	50 OF 50 (100.00%)
EXPLORE APPLICATION LAYER DoS ATTACKS	✓	45 OF 45 (100.00%)
MITIGATE SLOWLORIS ATTACKS	✓	45 OF 45 (100.00%)
EXPLORE DNS AMPLIFICATION ATTACKS		25 OF 45 (55.56%)

Second Attempt completed task 5

1 OF 5		9M 34s	45 OF 240 (18.75%)
OBJECTIVES COMPLETE		ELAPSED TIME	SCORE
WEDNESDAY, 18 SEPTEMBER 2024,		WEDNESDAY, 18 SEPTEMBER 2024,	
12:11 PM		12:21 PM	
START TIME		END TIME	
OBJECTIVE	COMPLETED	SCORE	
LAUNCH AND DETECT A TCP SYN FLOOD ATTACK		0 OF 55 (0.00%)	
MITIGATE A TCP SYN FLOOD ATTACK		0 OF 50 (0.00%)	
EXPLORE APPLICATION LAYER DoS ATTACKS		0 OF 45 (0.00%)	
MITIGATE SLOWLORIS ATTACKS		0 OF 45 (0.00%)	
EXPLORE DNS AMPLIFICATION ATTACKS	✓	45 OF 45 (100.00%)	

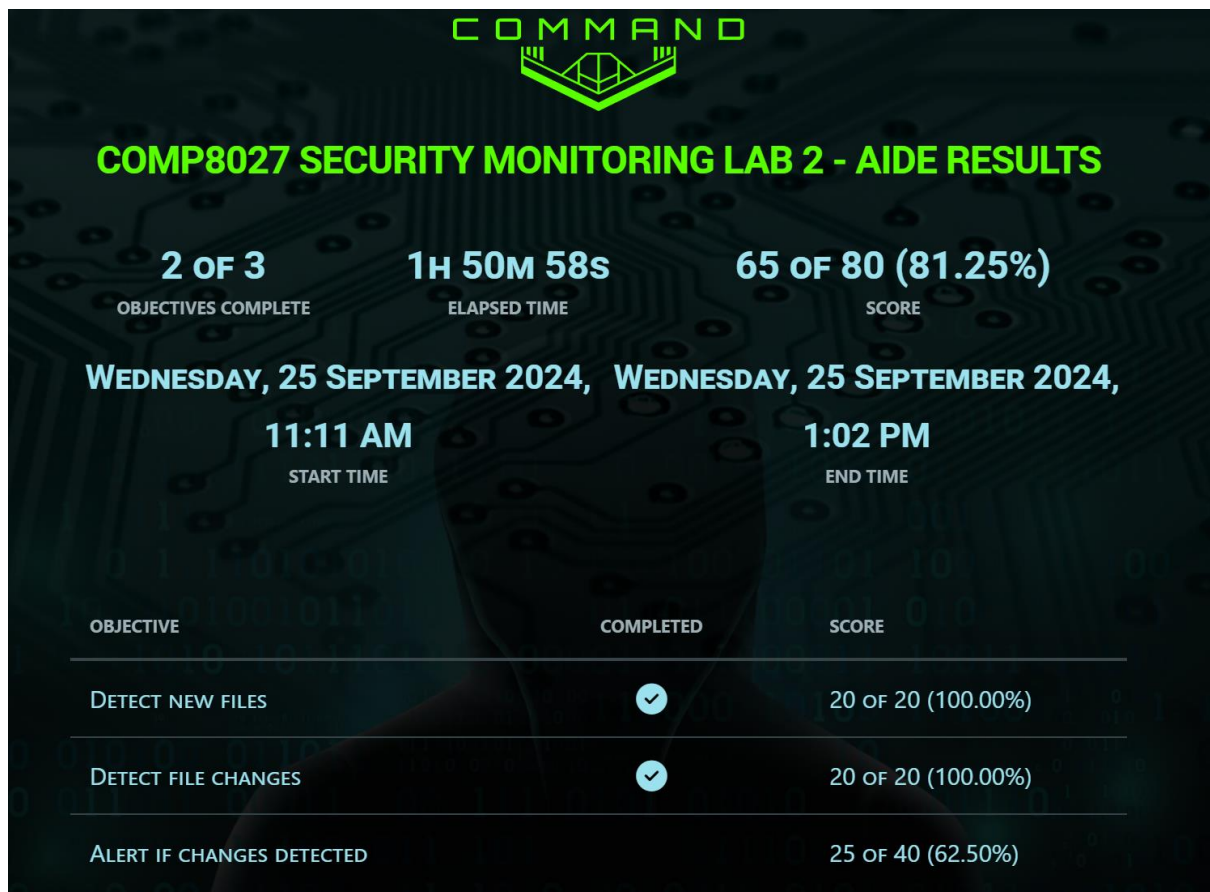
Lab 2

Task 4

1. AIDE successfully detected the creation of the new files /etc/etcFile1, /etc/etcFile2, /bin/binFile1, and /bin/binFile2. This confirms the integrity rules in aide.conf are functioning as intended, specifically for detecting newly created files.

Task 5

1. AIDE successfully detected changes in the contents of /etc/etcFile1 and /bin/binFile1, confirming the effectiveness of the MD5 checksum rule. Additionally, AIDE identified permission changes for /etc/etcFile2 and /bin/binFile2, reflecting the accuracy of the permission checks.
2. No changes were missed during this check, as both file modifications and permission changes were properly captured.



Task 6 crontab:

This wouldn't work on the vm but I have it working ill include the steps and screenshot to prove so

1. "Crontab -e" – here we enter and edit the cron job.
2. `"* * * * * sudo /usr/sbin/aide --config=/etc/aide/aide.conf --check | grep "found differences" && logger -t aide "AIDE detected changes" >> /var/log/aide_alert.log"` – here we add the command to schedule the aide check for every minute so we can verify if changes were detected
3. `"sudo nano /etc/rsyslog.d/00-aide.conf"` here we create our syslog file in the directory as shown.
4. `"if $msg contains 'AIDE detected changes' then /var/log/aide_alert.log & stop"` - Next what I did was created a condition which ensured that the log file only can be updated when the message contains "Aide detected changes" I have it so the logged changes are sent to the aide_alert.log directory in the /var/log directory and then the "&stop" so it stops processing the log.
5. `"sudo systemctl restart rsyslog"` we restart the syslog and then we wait for a minute in this case I waited 4 minutes and we then checked our alert directory to see if changes were detected using the command `"cat /var/log/aide_alert.log"` here is a screenshot of the output:

```

crontab: installing new crontab
cyrinuser@ubuntu20:~$ cat /var/log/aide_alert.log
Sep 17 22:55:01 ubuntu20 CRON[3629]: (root) CMD (sudo /usr/sbin/aide --confi
g=/etc/aide/aide.conf --check | grep "found differences" && logger -t aide "
AIDE detected changes" >> /var/log/aide_alert.log)
Sep 17 22:56:01 ubuntu20 CRON[3641]: (root) CMD (sudo /usr/sbin/aide --confi
g=/etc/aide/aide.conf --check | grep "found differences" && logger -t aide "
AIDE detected changes" >> /var/log/aide_alert.log)
Sep 17 22:57:01 ubuntu20 CRON[3646]: (root) CMD (sudo /usr/sbin/aide --confi
g=/etc/aide/aide.conf --check | grep "found differences" && logger -t aide "
AIDE detected changes" >> /var/log/aide_alert.log)
cyrinuser@ubuntu20:~$

```

Lab 3:

Log Analysis with RSYSLOG



LOG ANALYSIS WITH RSYSLOG (NM104) RESULTS

7 of 7 OBJECTIVES COMPLETE 38m 27s ELAPSED TIME 100 OF 100 (100.00%) SCORE

WEDNESDAY, 2 OCTOBER 2024, 11:11 AM

START TIME — CURRENTLY RUNNING

REFRESHING EVERY 30 SECONDS [RAISE UPDATED](#)

OBJECTIVE	COMPLETED	SCORE
CONFIGURE CENTRAL LOGGER	✓	15 of 15 (100.00%)
CONFIGURE LOG FORWARDING ON LINUX AND FREEBSD	✓	15 of 15 (100.00%)
CONFIGURE LOG FORWARDING ON WINDOWS 10	✓	15 of 15 (100.00%)
ROTATE LOG FILES	✓	10 of 10 (100.00%)
CREATE LOG FILTERS	✓	15 of 15 (100.00%)
CONFIGURE LOGWATCH	✓	15 of 15 (100.00%)
DEMONSTRATE FAIL2BAN	✓	15 of 15 (100.00%)

Lab 4:

```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32>
PS C:\Windows\system32> auditpol /get /category:*
system audit policy
category/Subcategory          Setting
system
  Security System Extension    No Auditing
  System Integrity             Success and Failure
  IPsec Driver                 No Auditing
  Other System Events          Success and Failure
  Security State Change        Success
logon/logoff
  Logon                        Success and Failure
  Logoff                       Success
  Account Lockout              Success
  IPsec Main Mode              No Auditing
  IPsec Quick Mode             No Auditing
  IPsec Extended Mode          No Auditing
  Special Logon                Success
  Other Logon/Logoff Events     No Auditing
  Network Policy Server        Success and Failure
  User / Device Claims         No Auditing
  Group Membership             No Auditing
object access
  File System                  No Auditing
  Registry                    No Auditing
  Kernel Object                No Auditing
  SAM                          No Auditing
  Certification Services       No Auditing
  Application Generated         No Auditing
  Handle Manipulation           No Auditing
  File Share                    No Auditing
  Filtering Platform Packet Drop No Auditing
  Filtering Platform Connection No Auditing
  Other Object Access Events    No Auditing
  Detailed File Share           No Auditing
  Removable Storage             No Auditing
  Central Policy Staging        No Auditing
privilege use
  Non Sensitive Privilege Use   No Auditing
  Other Privilege Use Events     No Auditing
  Sensitive Privilege Use       No Auditing
detailed tracking
  Process Creation              No Auditing
  Process Termination           No Auditing
```

```
DPAPI Activity                No Auditing
RPC Events                     No Auditing
Plug and Play Events          No Auditing
Token Right Adjusted Events    No Auditing
policy change
  Audit Policy Change          Success
  Authentication Policy Change Success
  Authorization Policy Change  No Auditing
  MPSSVC Rule-Level Policy Change No Auditing
  Filtering Platform Policy Change No Auditing
  Other Policy Change Events    No Auditing
account management
  Computer Account Management   No Auditing
  Security Group Management     Success
  Distribution Group Management No Auditing
  Application Group Management  No Auditing
  Other Account Management Events No Auditing
  User Account Management       Success
S Access
  Directory Service Access      No Auditing
  Directory Service Changes     No Auditing
  Directory Service Replication No Auditing
  Detailed Directory Service Replication No Auditing
account logon
  Kerberos Service Ticket Operations No Auditing
  Other Account Logon Events       No Auditing
  Kerberos Authentication Service  No Auditing
  Credential Validation           No Auditing
PS C:\Windows\system32>
```

COMP8027 Security Monitoring Lab 4 - Windows Event Logs



COMP8027 SECURITY MONITORING LAB 4 - WINDOWS EVENT LOGS RESULTS

Run: WEDNESDAY, 9 OCTOBER 2024, 11:05 AM

12 of 12
OBJECTIVES COMPLETE

1h 56m 45s
ELAPSED TIME

100 of 100 (100.00%)
SCORE

WEDNESDAY, 9 OCTOBER 2024, 11:05 AM
START TIME

WEDNESDAY, 9 OCTOBER 2024, 1:43 PM
END TIME

OBJECTIVE	COMPLETED	SCORE
CONFIGURE AUDIT POLICY	☒	20 of 20 (100.00%)
1. WHEN USER ALICE WAS CREATED	☒	7 of 7 (100.00%)
2. WHEN ALICE'S PASSWORD WAS CHANGED BY ALICE	☒	7 of 7 (100.00%)
3. WHEN ALICE'S PASSWORD WAS CHANGED BY AN ADMINISTRATOR	☒	7 of 7 (100.00%)